NONNESTED EXTREMAL TARGETS IN A RATIONAL RADIX–MÖBIUS SEMIGROUP

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Abstract. We study a noncommutative semigroup of rational interval maps generated by the integer-radix transformation $T_B(x) = \{Bx\}$ and the Möbius-conjugate branch $S_B = H \circ T_B \circ H^{-1}$, where $H(x) = (1 + x)^{-1}$. For the decimal benchmark $B = 10$, seed $17/19$, and a finite certified terminal set, we define extremal target sets

$$P_m(10, \kappa) = \{x : q(x) > \kappa 10^m\}, \qquad q(a/b) = b/(b - a).$$

Exact rational enumeration gives the first-hit times

$$(\tau_1, \tau_2, \tau_3, \tau_4) = (45, 85, 120, 162)$$

at $\kappa = 11$. The corresponding first-hit states are not nested Bellman optima: optimally continuing an earliest depth-m state reaches the next target strictly later than the global depth-$(m + 1)$ optimum, with delays 2, 19, and 6 actions in three consecutive transitions. This phenomenon is robust, not threshold-fine-tuned: all four hitting times remain unchanged for

$$\kappa \in [10.6197, 11.643),$$

and all three nonnesting inequalities persist simultaneously for

$$\kappa \in [10.6197, 11.01).$$

The results give exact arithmetic counterexamples to current target height and attained target depth as sufficient Bellman variables. We describe the certified computation, threshold atlas, and evidence boundary, and formulate the resulting hidden-state problem independently of any unresolved depth-five optimum.

1. Introduction

Integer-radix transformations of the form $x \mapsto \{Bx\}$ are classical objects in the theory of expansions and interval dynamics [1, 2, 3]. Möbius and Farey-type maps, their natural extensions, and their symbolic and geodesic codings form another mature line of research [4, 5, 6, 8, 9, 10]. Recent work also emphasizes finite rational orbits in generalized continued-fraction systems and rational semigroup actions [11, 12], while new natural-extension constructions continue to broaden the class of projective arithmetic maps [13, 14, 15]. The present work studies a concrete coupling of these structures in which an integer-radix map and a Möbius-conjugate branch act on primitive rationals, with exact reduction after every action.

Our focus is not the introduction of a new radix map or a new version of Bellman's principle. Rather, we isolate an exact arithmetic phenomenon that arises from their interaction: the state that first reaches one extremal target need not be the best precursor for the next, deeper target. In the benchmark system considered here, three consecutive target transitions exhibit this failure of nesting. The effect is large: states with present height of order one can outperform already attained portals with heights of order 10^3 or 10^4.

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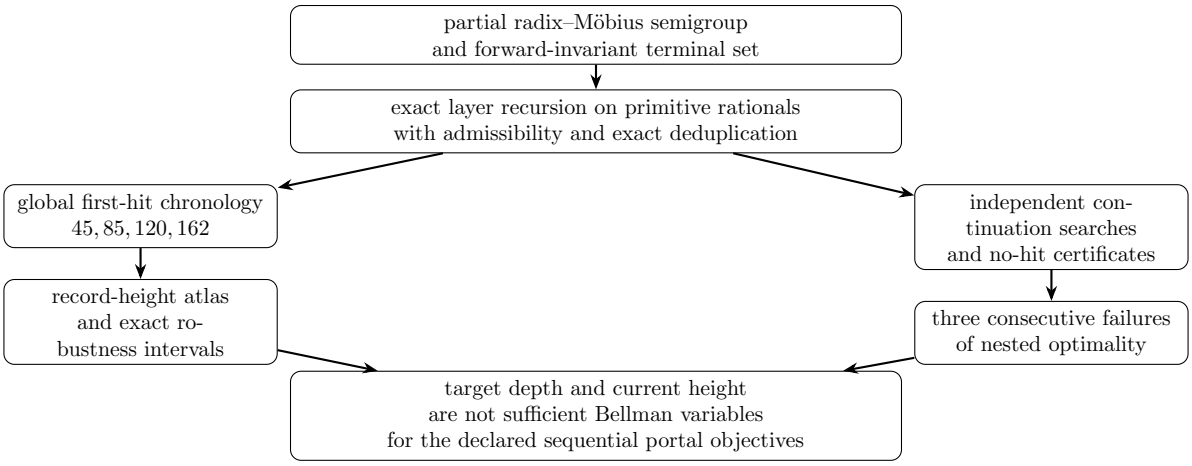


FIGURE 1. Logical dependency graph of Paper I. Symbolic definitions and the certified terminal convention feed an exact finite recursion; global layers and independent continuation searches establish chronology, nonnesting and robustness before the Bellman consequences are drawn.

The principle of optimality requires a state description that retains all future-relevant information [20, 21]. State augmentation for non-Markovian objectives is well known in control and planning [22, 23]. What is specific here is an exact arithmetic realization: target depth and present height fail as sufficient variables, while the full rational state remains Markovian. A companion line of work identifies additional Farey and p-adic coordinates that organize the hidden future geometry; those coordinates are not required for the main theorems of this paper.

The contributions are:

(1) a base-B rational radix–Möbius semigroup with explicit branch matrices;
(2) exact first-hit times and first-hit sets for four extremal target depths in the decimal benchmark;
(3) exact nonnesting certificates for three consecutive target transitions;
(4) an exact threshold-robustness atlas proving that the phenomenon persists on a positive interval of κ;
(5) a reproducible computer-assisted proof protocol based entirely on primitive integer arithmetic.

The depth-five hitting time is not needed here. Its current exact interval is $186 \le \tau_5 \le 212$, but none of the claims below depends on closing that interval.

2. The rational radix–Möbius semigroup

Fix an integer $B \ge 2$. Define

$$T_B(x) = \{Bx\}, \qquad H(x) = \frac{1}{1+x}.$$

We regard T_B as a self-map of $[0,1]$ (with $T_B(1) = 0$) and define the conjugate generator

$$S_B = H \circ T_B \circ H^{-1}$$

as a partial transformation with natural domain

$$D_S = H([0,1]) = [1/2, 1].$$

Thus the generated object is a semigroup of partial rational transformations; a word is admissible at a state only when every partial generator occurring in it is defined. If $x = a/b$ is reduced, then

the active branch digits are

$$d_T = \left\lfloor \frac{Ba}{b} \right\rfloor, \qquad d_S = \left\lfloor \frac{B(b-a)}{a} \right\rfloor.$$

Proposition 2.1 (Branch formulae)**.** *For a reduced rational a/b,*

$$T_B(a/b) = \frac{Ba - d_T b}{b}$$

before primitive reduction. Whenever S_B is admissible,

$$S_B(a/b) = \frac{a}{Bb - (B - 1 + d_S)a}$$

before primitive reduction. The corresponding integer matrices are

$$M_T(d) = \begin{pmatrix} B & -d \\ 0 & 1 \end{pmatrix}, \qquad M_S(d) = \begin{pmatrix} 1 & 0 \\ -(B-1+d) & B \end{pmatrix},$$

and both have determinant B.

The two generators need not commute. In the decimal benchmark,

$$T(S(17/19)) = 1/2, \qquad S(T(17/19)) = 9/14.$$

Thus the action is genuinely noncommutative.

2.1. **Benchmark and terminal set.** For the remainder of the paper we set $B = 10$ and use the seed

$$x_0 = 17/19.$$

We remove the following certified forward-invariant terminal set:

$$\begin{aligned} E = \{&0, 1, 1/2, 1/3, 2/3, 3/4, 4/5, 5/6, 7/8, 7/9, 7/10, \\ &2/11, 4/11, 7/11, 9/11, 11/20, 7/12, 2/13, 5/13, 6/13, 7/13, 8/13, 11/13\}. \end{aligned}$$

Direct child auditing shows that every admissible child of every element of E remains in E. The accompanying terminal-set audit records all 23 states and all 46 attempted T/S branches (admissible or not), with zero admissible escapes from E.

3. Extremal target sets

For $x = a/b < 1$ reduced, define the extremal height

$$q(x) = \frac{1}{1-x} = \frac{b}{b-a}.$$

For $m \geq 1$ and $\kappa > 0$, define

$$P_m(10, \kappa) = \{x : q(x) > \kappa 10^m\}.$$

Let $\tau_m(\kappa)$ be the earliest time at which a nonterminal trajectory from $17/19$ reaches $P_m(10, \kappa)$.

The strict inequality in the target definition matters at robustness boundaries. If the threshold equals the largest previous height, that previous state is not yet a hit.

4. Exact portal chronology

Theorem 4.1 (Exact benchmark first hits)**.** *At $\kappa = 11$,*

$$\boxed{(\tau_1, \tau_2, \tau_3, \tau_4) = (45, 85, 120, 162).}$$

The first-hit sets are:

m	τ_m	*first-hit states*
1	*45*	128/129
2	*85*	1202/1203
3	*120*	11156/11157, 11532/11533, 11642/11643
4	*162*	115482/115483, 124852/124853

Computer-assisted proof. Every admissible T and S child was generated in exact primitive integer arithmetic. The set E was removed only after its forward invariance had been independently checked. States were deduplicated exactly within each layer. At depth 162 the exact layer contains 329,106,757 states. The maximum heights of all preceding layers are below the corresponding target thresholds, while the displayed states exceed them at the stated first-hit layers. The layer maxima and counts are included in the supplementary threshold atlas. □

5. Nonnested first-hit states

For a state x reached at time t, let $\sigma_m(x)$ denote the least number of additional actions required to reach P_m, with ∞ if no such continuation exists.

Theorem 5.1 (Three consecutive failures of nesting)**.** *At $\kappa = 11$:*

$$\begin{aligned} 45 + \sigma_2(128/129) &= 87 > 85 = \tau_2, \\ 85 + \sigma_3(1202/1203) &= 139 > 120 = \tau_3, \\ 120 + \sigma_4(11156/11157) &= 168 > 162 = \tau_4. \end{aligned}$$

Moreover, the other two first-hit states at depth three, 11532/11533 *and* 11642/11643*, have no depth-four hit by time* 162*.*

Proof. Independent exact target searches give

$$128/129 \xrightarrow[42 \text{ actions}]{} 1100/1101,$$

$$1202/1203 \xrightarrow[54 \text{ actions}]{} 13542/13543,$$

and

$$11156/11157 \xrightarrow[48 \text{ actions}]{} 122088/122089.$$

The searches enumerate 55,144, 3,955,616, and 30,380,076 distinct states, respectively, before the first hit. Separate exact searches from 11532/11533 and 11642/11643 find no hit through total time 162. □

This proves that globally earliest targets do not form an optimal nested chain. The phenomenon is not merely that one optimal path is nonunique; the target-optimal state itself can be future-suboptimal.

Corollary 5.2 (Target depth is not a sufficient Bellman state)**.** *No dynamic program whose state records only elapsed time and attained target depth can satisfy the principle of optimality for this system.*

Corollary 5.3 (Current height is not a Bellman value)**.** *The scalar $q(x)$ cannot order states by future portal value. The following exact same-time comparisons reverse the ordering induced by present height:*

checkpoint	*target-optimal state*	q	*latent winning state (q)*	*remaining costs*
45	128/129	129	103/131 (131/28)	$42 > 40$
85	1202/1203	1203	956/1573 (1573/617)	$54 > 35$
120	11156/11157	11157	11803/22160 (22160/10357)	$48 > 42$

Thus states of much smaller current height can have strictly smaller exact cost to the next portal.

6. Threshold robustness

Let

$$M_d = \max\{q(x) : x \text{ is reachable at depth } d\}.$$

The function $\tau_m(\kappa)$ is piecewise constant at finite horizon: it can change only when $\kappa = M_d/10^m$ for a record height M_d.

Proposition 6.1 (First-hit interval)**.** *If d is a record time for M_d and $M_{<d} = \max_{j<d} M_j$, then*

$$\tau_m(\kappa) = d$$

for

$$\kappa \in \left[\frac{M_{<d}}{10^m}, \frac{M_d}{10^m}\right).$$

Theorem 6.2 (Simultaneous robustness of the benchmark chronology)**.** *The vector*

$$(\tau_1, \tau_2, \tau_3, \tau_4) = (45, 85, 120, 162)$$

remains unchanged for every

$$\boxed{\kappa \in \left[\frac{106197}{10000}, \frac{11643}{1000}\right) = [10.6197, 11.643).}$$

The individual intervals are:

m	τ_m	exact κ interval
1	45	$[93/10, 129/10)$
2	85	$[903/100, 1203/100)$
3	120	$[10019/1000, 11643/1000)$
4	162	$[106197/10000, 124853/10000)$

Theorem 6.3 (Robust nonnesting interval)**.** *All three nonnesting inequalities of Theorem 5.1 hold simultaneously for*

$$\boxed{\kappa \in \left[\frac{106197}{10000}, \frac{1101}{100}\right) = [10.6197, 11.01).}$$

Within this interval all three depth-three first-hit states remain optimal at time 120*, both depth-four first-hit states remain optimal at time* 162*, and none of the three depth-three optima reaches depth four by time* 162*.*

Proof. The exact continuation stability intervals are

$$[10.33, 11.01), \qquad [8.827, 13.543), \qquad [4.4853, 12.2089),$$

for the three displayed continuations. Intersecting each with the corresponding global first-hit interval and then intersecting the three results gives $[10.6197, 11.01)$. The maximum heights reached

by the other two depth-three optima through time 162 are 44,400 and 28,408, so they cannot reach the depth-four target anywhere in this interval. □

The positive-width interval rules out the interpretation that nonnesting is an artifact of the particular calibration $\kappa = 11$.

7. Certified computational method

Proposition 7.1 (Exactness of the layer recursion)**.** *Let $\mathcal{L}_0 = \{x_0\} \setminus E$ and define recursively*

$$\mathcal{L}_{n+1} = \{A(x) \notin E : x \in \mathcal{L}_n,\ A \in \{T, S\},\ A \text{ is admissible at } x\},$$

where every rational is reduced to primitive form and duplicate states are identified. Then $\mathcal{L}_n$ is exactly the set of nonterminal states reachable from x_0 by an admissible word of length n. Consequently,

$$\tau_m(\kappa) = \min\{n : \mathcal{L}_n \cap P_m(10, \kappa) \neq \varnothing\}.$$

Proof. The statement is immediate at $n = 0$. Assume it holds at time n. Every admissible word of length $n+1$ has an admissible prefix of length n and a final generator $A \in \{T, S\}$, hence its endpoint is generated by the displayed recursion. Conversely, every state produced by the recursion extends an admissible length-n word by one admissible generator. Removing E loses no later nonterminal trajectory because E is forward invariant. Primitive reduction and exact deduplication change only the representation and multiplicity of a rational state, not reachability. Induction proves the equality of layers, and the first-hit formula follows. □

All computations use reduced pairs of nonnegative integers. No floating-point comparison decides target membership. Target membership is tested on every exact child before any certified terminal or viability pruning. The proof stack contains:

(1) exact branch generation;
(2) exact terminal-set removal;
(3) exact state deduplication;
(4) independent target searches for continuation costs;
(5) canonical CSV ledgers of layer counts and maxima;
(6) independent verifiers and cryptographic hashes in the larger project archive.

The repository contains the exact layer-maxima ledger through depth 162, the threshold atlas, a compact benchmark receipt, the complete terminal-set audit, five continuation profiles, replay source, an independent receipt checker, and cryptographic hashes. The lightweight reviewer bundle includes the compact evidence needed to audit the headline claims. The complete 329,106,757-state depth-162 payload is not duplicated in ordinary Git; clean archival deposition or independent regeneration remains a pre-DOI gate. The result is therefore a computer-assisted theorem with a finite and explicitly delimited evidence chain, rather than a numerical simulation.

8. Relation to previous work

The radix generator belongs to the classical theory of β-transformations and expansions [1, 2, 3]. Farey and continued-fraction systems, their natural extensions, induced transformations, and their geodesic codings are developed in [4, 5, 6, 7, 8, 9, 10, 13]. Recent finite-orbit results show that rational trajectories can encode nontrivial statistical information in generalized continued-fraction systems [11], and recent work on rational semigroup actions gives structurally nearby examples of periodic and transitive arithmetic dynamics [12]. Arithmetic dynamics supplies the broader setting for rational iteration [18, 19].

The need for sufficient state in dynamic programming is classical [20, 21]; automata-based state augmentation for non-Markovian objectives appears, for example, in [22]. Likewise, invariant or survivor sets in open expanding systems are established objects [24], and independently checkable

certificates are an active topic in formal reachability verification [25]. We therefore do not claim novelty for radix dynamics, projective/Farey methods, state augmentation, trapping, or certification as general concepts.

The claim here is narrower. After comparison with these nearby literatures, we are not aware of a result for this exact rational radix–Möbius system that simultaneously provides (i) certified extremal first-hit times, (ii) explicit nonnesting across consecutive extremal targets, and (iii) a positive exact threshold interval on which that nonnesting persists. This is a provisional priority statement rather than a proof of historical uniqueness; the repository contains a claim-by-claim literature matrix, and specialist review remains part of the pre-submission validation plan.

9. Discussion

The examples distinguish three notions that coincide in simpler stopping problems:

(1) being the earliest state to cross the current target;
(2) maximizing the current observable height;
(3) minimizing the time to a deeper target.

They are different in the present semigroup.

The full rational state is Markovian, but compressed observables can discard decisive arithmetic phase information. Companion results show that Farey-neighbor and p-adic coordinates can select different controls from the same visible rational; those results will be treated separately to avoid conflating the nonnesting theorem with a proposed minimal hidden-state representation.

Several questions remain open. Does nonnesting persist for infinite families of bases, seeds, and target parameters? Can one bound the number of target-optimal states? What arithmetic invariants predict a latent winner? How do finite modular traps divide the state space into escape and nonescape basins? These questions are independent of the unresolved exact value of τ_5.

10. Limitations and evidence boundary

The exact threshold atlas in this paper is complete through depth 162 for the benchmark seed, base, and terminal convention. The robustness theorems are exact on their stated intervals, but no universality in B, seed, or κ is claimed. The hidden-state interpretation is supported by exact counterexamples; minimality of any particular augmented coordinate system is not proved here. The compact theorem stack is internally replayable, while archival deposition or clean regeneration of the full depth-162 layer remains open. No external clean-room implementation has yet reproduced the complete chronology. Finally, the priority claim remains provisional pending closed-database citation chaining and external specialist review.

11. Data and code availability

The public research repository is available as the RGΦ project repository. The submission package contains the compact global layer profile, exact continuation profiles, threshold-atlas data and generator, terminal-set audit, independent receipt checker, and SHA-256 manifests. The complete depth-162 state payload is outside the lightweight package and is an explicit archival/clean-regeneration gate. A public archival DOI and Software Heritage identifier will be assigned only after the reviewed commit and evidence set are frozen; no identifier is claimed before deposition.

Appendix A. Exact evidence tables

The principal nonnesting table is:

source	source time	global target time	continued arrival	delay
128/129	45	85	87	2
1202/1203	85	120	139	19
11156/11157	120	162	168	6

The common exact robustness interval is

$$[10.6197, 11.01),$$

not a rounded confidence interval. Its endpoints are rational critical values obtained from exact layer maxima.

References

[1] A. Rényi, *Representations for real numbers and their ergodic properties*, Acta Math. Acad. Sci. Hungar. (1957), doi:10.1007/BF02020331.

[2] W. Parry, *On the β-expansions of real numbers*, Acta Math. Acad. Sci. Hungar. (1960), doi:10.1007/BF02020954.

[3] K. Dajani and C. Kraaikamp, *Ergodic Theory of Numbers*, Mathematical Association of America, 2002.

[4] H. Nakada, *Metrical theory for a class of continued fraction transformations and their natural extensions*, Tokyo J. Math. (1981).

[5] C. Series, *The modular surface and continued fractions*, J. London Math. Soc. (1985).

[6] S. Isola, *On the spectrum of Farey and Gauss maps*, arXiv:math/0308017 (2003).

[7] C. Merriman, *Geodesic flows and slow downs of continued fraction maps*, arXiv:2307.11700 (2023).

[8] G. Panti, *There is only one Farey map*, Ann. Mat. Pura Appl. (2026), doi:10.1007/s10231-026-01693-1; arXiv:2506.02984.

[9] T. A. Schmidt, *First return systems for some continued fraction maps*, arXiv:2511.12835 (2025).

[10] K. Dajani, C. Kraaikamp, and S. Sanderson, *Inducing contractions of the mother of all continued fractions*, J. Number Theory 278 (2026), 816–874, doi:10.1016/j.jnt.2025.05.015.

[11] V. Berthé, S. Cantrell, J. Lee, and M. Pollicott, *Multidimensional statistics for finite orbits of generalised continued fractions*, arXiv:2507.11295 (2025).

[12] M. Javaheri, *Transitive actions and purely periodic N-expansions*, J. Number Theory 283 (2026), doi:10.1016/j.jnt.2025.11.013.

[13] Y. Chen, K. Dajani, Y. Hu, and C. Kraaikamp, *A strange continued fraction associated with the Romik map*, arXiv:2605.21098 (2026).

[14] J. Fox and T. Garrity, *The natural extension for the triangle map (a multi-dimensional continued fraction) with an internal symmetry from Young conjugation*, arXiv:2606.28014 (2026).

[15] H. Nakada, R. Natsui, and J. Thuswaldner, *Farey graphs and geodesic expansions of complex continued fractions*, arXiv:2603.28468 (2026).

[16] L. Peng, *On the hitting depth in the dynamical system of continued fractions*, Chaos Solitons Fractals 69 (2014), 22–30, doi:10.1016/j.chaos.2014.09.003.

[17] N. Langeveld and D. Ralston, *On a tree of rational functions related to continued fractions*, Indag. Math. (N.S.) (2026), doi:10.1016/j.indag.2026.06.001; arXiv:2512.11595.

[18] J. H. Silverman, *The Arithmetic of Dynamical Systems*, Graduate Texts in Mathematics 241, Springer, 2007.

[19] V. Anashin and A. Khrennikov, *Applied Algebraic Dynamics*, De Gruyter, 2009.

[20] R. Bellman, *On the theory of dynamic programming*, Proc. Natl. Acad. Sci. USA 38 (1952), 716–719.

[21] B. Light, *The principle of optimality in dynamic programming: a pedagogical note*, arXiv:2302.08467 (2023).

[22] R. Brafman, G. De Giacomo, and F. Patrizi, *Specifying non-Markovian rewards in MDPs using LDL on finite traces*, arXiv:1706.08100 (2017).

[23] M. Jones and M. Peet, *A generalization of Bellman's equation with application to path planning, obstacle avoidance and invariant set estimation*, Automatica / arXiv:2006.08175 (2021).

[24] D. Kong, B. Sun, and Z. Wang, *Open dynamical systems with a moving hole*, arXiv:2505.02336 (2025).

[25] K. Chatterjee, T. Quatmann, M. Schäffeler, M. Weininger, T. Winkler, and D. Zilken, *Fixed point certificates for reachability and expected rewards in MDPs*, arXiv:2501.11467 (2025).